

Toby McGuire

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SKILLS

Languages: C, C++, C#, Java, Python, TypeScript

Frameworks & Tools: React, Node.js, ROS, Docker, Git, Linux, AWS (Cognito, S3), Unity, Unreal Engine, Visual Studio, VS Code, Blender

Machine Learning & AI: TensorFlow, YOLOv5, NumPy, computer vision pipelines, synthetic data generation

Data & Storage: PostgreSQL, REST APIs, JWT authentication

CERTIFICATIONS

Secret Security Clearance

Active

AWS Certified Cloud Practitioner

February 2025

EXPERIENCE

The Boeing Company

Florida

Software Engineer IC2

July 2023 - Present

- Developed automated full-level integration tests across multiple flight and space programs (SLS, MQ-25, F/A-18), ensuring vehicle software met mission-critical project requirements and reducing manual verification efforts.
- Designed custom automation tooling that eliminated months of repetitive engineering work and cut rework by over 50%.
- Selected by leadership to join multiple high-priority cross-functional teams, contributing across subsystems and infrastructure code with direct impact on program delivery.

Nauticus Robotics

Houston, Texas

Software Engineer Intern

May 2022 - August 2022

- Built a computer vision data generation pipeline in Blender, producing 30,000+ labeled training samples for underwater object detection using YOLOv5 and TensorFlow, replacing manual labeling entirely.
- Automated stereo image processing pipeline, improving batch throughput by 90% and enabling overnight conversion of stereo image pairs into 3D point cloud models.
- Implemented conic reconstruction algorithms from research, applying computer vision techniques in the real-world.

Ohio State Underwater Robotics

Columbus, Ohio

Software Engineer

August 2020 - May 2024

- Built a Unity-based simulation that ingested real-time sensor data from onboard computers to visualize navigation behaviors and validate perception algorithms using ROS pub/sub.

PROJECTS

Saucer - Recipe Aggregator App

2024 - Present

Creator, Full-Stack Developer

- Built a production full-stack web application using **TypeScript, React, and Node.js** with user authentication (AWS Cognito + JWT), cloud storage (AWS S3), and real-time collaborative features.
- Integrated AI-assisted recipe parsing and an OCR pipeline to extract and structure recipe data from uploaded files.
- Architected backend REST API with PostgreSQL and deployed via Docker and AWS, supporting multi-user accounts.

SUMO - Multiplayer Steam Game

2022 - 2024

Creator, Lead Developer

- Shipped a live multiplayer party game on Steam using **Unity and C#** with Steamworks SDK for player authentication and session management, and FishNetworking for authoritative server-side netcode.
- Implemented encrypted persistent data storage using serialization and data salting to protect player save data.
- Contributed bug fixes and features back to FishNetworking, the **open-source** multiplayer framework powering the game.

EDUCATION

The Ohio State University, College of Engineering

Columbus, Ohio

B.S. in Computer Science and Engineering | GPA: 3.84 | Specialization in Artificial Intelligence

May 2023